Derivative Securities FNCE 30007

Tutorial 3

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Review

If there are no transaction costs, taxes, or differences in interest rates, arbitrageurs can exploit any price discrepancies between identical assets. Their actions would push the prices to equalize, ensuring that the law of one price (LOP) holds.

Let's consider the simplest case: domestic asset with no dividend/coupon payment, no storage cost, no convenience yield.

- During the tutorial, we will face more challenging questions, but the idea is based on this simple case).

Review

Notation:

- S_0: The spot price of the underlying asset as of today.
- F_0: The futures or forward price as of today.
- T: Time to maturity (of the contract)
- r: Risk-free interest rate (continuously compounding)

The relationship between F_0 and S_0 is:

$$F_0 = S_0 e^{rT}$$

.

- Borrow money at r then buy and hold the asset until the maturity date (maturity date of the forward contract) = Long the forward contract.
- We will pay the same for these two activities.

Review

Example:

- Spot price of gold is US$600
- 1-year futures price of gold is US$650
- 1-year US$ interest rate is 5% p.a.
- No income or storage costs for gold
- $F_0 = 600 \times e^{0.05 \times 1} = 630.76$
- Futures are **overvalued** at $650 (Buy low sell high)

	$t = 0$	$t = T$
Borrow	+$600	−$630.76 (repay loan)
Purchase gold	−$600	—
Short futures	—	Sell gold at T for +$650
Net cash flow	$0	+$19.24 (risk-free)

Review

Example:

- All else equal to previous example except that 1-year futures price of gold is US$590

$$F_0 = 600 \times e^{0.05 \times 1} = 630.76$$

Futures are **undervalued** at $590 (buy futures), sell spot.

	$t = 0$	$t = T$
Borrow gold and sell	+$600	
Invest at risk-free	-$600	+$630.76 (receive investment)
Enter futures at $590	$0	-$590 (buy gold, return it)
Net Cash Flow	$0	+$40.76 (risk-free)

Review

Other more challanging cases:

- Asset with known (discrete) income during the life of the futures/forward contract:

  $$F_0 = (S_0 - I)e^{rT}$$

  where I is the present value of income during the life of the forward contract.

- Asset with know yields (use d to represent the average annualized continuously compounding yield on the asset):

  $$F_0 = S_0 e^{(r-d)T}$$

- Foreign asset can be viewed as an asset with special "dividend yield" r_f. Note this r_f is the risk-free rate in the foreign country.

Review

- Consumption asset with storage cost and convience yield:

$$F_0 e^{yT} = (S_0 + U)e^{rT}$$

  - U is the PV of all storage costs during the life of the contract
  - y is the convenience yield which express the additional value that direct ownership of the asset entails as an annualized c.c. rate.

- If u is expressed as storage costs as a proportion of spot price, then we have

$$F_0 e^{yT} = S_0 e^{(r+u)T}$$

- Remove the convenience yield, the previous = becomes to $\leq$

Review

Valuing an existing contract:

- Suppose the F_0 is forward price that would apply to the contract today.
- K is delivery price in a forward contract entered into previously.
- The value of a long forward contract is: $f = (F_0 - K)e^{-rT}$
  - Since you are able to buy the underlying asset at K. Now you can sell the underlying asset at the same expiration date at F_0, you can get $F_0 - K$ at the expiration date of the contract to get a certain amount of money. Then the PV is f, the value of the forward contract as of today.
- Value of a short forward contract is: $f = (K - F_0)e^{-rT}$

Tutorial Question

Q1: A one-year long forward contract on a non-dividend paying stock is entered into when the stock price is \$40, and the risk-free rate of interest is 10% per annum with continuous compounding. Assume all conditions assumed in lectures hold.

(a) What is the forward price and the initial value of the forward contract?

(b) Six months later, the price of the stock is \$45, and the risk-free interest rate is still 10%. What is the forward price and the value of the forward contract?

The forward price in part (a) is $F = 40 \times e^{0.1} = 44.21$. The initial value of the contract is zero.

In part (b), the new forward price is $F = 45 \times e^{0.1/2} = 47.31$. The value of the forward in part (a) is

$$f = (47.31 - 44.21) \times e^{-0.1/2} = 2.95.$$

Tutorial Question

Q2: A stock is expected to pay a dividend of AUD 1 per share in two months and in five months. The stock price is AUD 50, and the risk-free rate of interest is 8% per annum with c.c. for all maturities. An investor has just taken a short position in a six-month forward contract on the stock.

(a) What is the forward price and the initial value of the forward contract?

(b) Three months later, the price of the stock is AUD 48, and the risk-free rate of interest is still 8% per annum. What is the forward price and the value of the short position in the forward contract?

Tutorial Question

Q2 Cont'd:
The PV of the cash flows using c.c. risk free rate: $I = 1 * e^{-0.08*\frac{2}{12}} + 1 * e^{-0.08*\frac{5}{12}} = 1.954$.

The forward price in part (a) is then:

$$F_0 = (50 - 1.954) \times e^{0.08/2} = 50.01.$$

with initial value 0. The forward price and the spot price are very close in this case because the dividend yield is equal to the risk-free rate (with small differences due to differences in compounding frequency).

Tutorial Question

In part (b) there remains only one dividend payment with a PV of:

$$I = 1 \times e^{-0.08(2/12)} = 0.9868.$$

The no-arbitrage forward price is now:

$$F = (48 - 0.9868) \times e^{0.08(3/12)} = 47.96.$$

and the value of short position in the forward contract is:

$$f = (K - F_0)e^{-rT} = (50.01 - 47.96) \times e^{-0.08(3/12)} = 2.01.$$

Note that because we are valuing a short position, the market value is the market value of an equivalent long forward (as derived in lectures) times negative one.

Tutorial Question

Q3: Suppose that the risk-free rate is 10% per annum and that the dividend yield on a stock index is 4% per annum. Both rates are expressed with continuous compounding. The index value is currently 400 and the futures price for a contract deliverable in four months is 405. What arbitrage opportunities does this create?

Tutorial Question

Q3 Cont'd: The no-arbitrage futures price is:

$$F_0 = S_0 e^{(r-d)T} = 400 \times e^{(0.1-0.04)(4/12)} = 408.08.$$

The futures price is too low and so the arbitrageur should short the stocks (or ETF) and invest the proceeds at the risk-free rate while taking a long futures position. At expiry, the arbitrageur either

(i) Deliverable: pays 405 in the futures market and takes delivery of the stocks in the index these uses these to cover the short, or

(ii) Cash settled: buys back the stocks (or ETF) in the spot market (cost S_T) to cover the short while recording an equivalent PnL from the cash-settled futures expiry (receive $S_T - 405$).

The arbitrageur will earn a profit of 3.08 at expiry ignoring things like transaction costs and shorting fees.

Tutorial Question

Q4: The two-month interest rates in Switzerland and the United States are 2% and 5% per annum, respectively, with continuous compounding. The spot price of the Swiss franc is \$0.80. The futures price for a contract deliverable in two months is \$0.81. What arbitrage opportunities does this create?

USD is now the domestic currency. The no-arbitrage futures price is:

$$F_0 = S_0 e^{(r-r^f)T} = 0.8 \times e^{(0.05-0.02)(2/12)} = 0.8040$$

The futures price is too high and so the arbitrageur should take a short position in the futures while simultaneously borrowing USD to fund an equivalent long spot position in the Swiss franc. At expiry, they deliver the Swiss franc from the spot market to the long side of the futures and earn a profit of \$0.81 - \$0.804 =\$0.006 per Swiss franc.

Tutorial Question

Q5: Suppose that F_1 and F_2 are two futures contracts on the same asset with times to maturity, t_1 and t_2, where $t_2 > t_1$. Prove that

$$F_2 \leq F_1 e^{r(t_2 - t_1)}$$

where r is the continuously compounded annualized interest rate (assumed constant) and there are no storage costs or income. For the purposes of this problem, assume that a futures contract is the same as a forward contract.

Tutorial Question

Q5 Cont'd: Prove this by contradiction. Suppose $F_2 > F_1 e^{r(t_2-t_1)}$. If this were the case, an arbitrageur could make a riskless profit by:

1. Going long a t_1 forward for 1 unit of the asset at a price of F_1.
2. Going short a t_2 forward for 1 unit of the asset at a price of F_2.
3. Borrowing F_1 at the risk-free rate at t_1 maturing at t_2 with a terminal value of $F_1 e^{r(t_2-t_1)}$.

At t_2 the arbitrageur delivers the asset obtained in the first forward to the long side of the second forward, receives F_2 and pays the terminal value of the loan in step 3. The PnL of this position is:

$$\text{PnL} = F_2 - F_1 e^{r(t_2-t_1)}$$

which would be greater than zero. Since arbitrage opportunities are quickly eliminated in liquid markets, it must be that:

$$F_2 \leq F_1 e^{r(t_2-t_1)}$$

Tutorial Question

Q6:When a known future cash outflow in a foreign currency is hedged by a company using a forward contract, there is no foreign exchange risk. When it is hedged using futures contracts, the daily settlement process does leave the company exposed to some risk. Explain the nature of this risk. In particular, consider whether the company is better off using a futures contract or a forward contract when

(a) The value of the foreign currency falls rapidly during the life of the contract

(b) The value of the foreign currency rises rapidly during the life of the contract

(c) The value of the foreign currency first rises and then falls back to its initial value

(d) The value of the foreign currency first falls and then rises back to its initial value

Assume that the forward price equals the futures price.

Tutorial Question

Q6 Cont'd: At expiry, the final gain or loss under a futures contract is equal to the gain or loss under the corresponding forward contract. However, the timing of the cash flows is different. When the time value of money is taken into account, a futures contract may prove to be more valuable or less valuable than a forward contract. The long forward contract provides a perfect hedge. The long futures contract provides a slightly imperfect hedge.

(a) In this case, the forward contract would lead to a slightly better outcome. The company will make a loss on its hedge. If the hedge is with a forward contract, the whole of the loss will be realized at the end. If it is with a futures contract, the loss will be realized day by day throughout the contract. On a present value basis the former is preferable.

Tutorial Question

(b) In this case, the futures contract would lead to a slightly better outcome. The company will make a gain on the hedge. If the hedge is with a forward contract, the gain will be realized at the end. If it is with a futures contract, the gain will be realized day by day throughout the life of the contract. On a present value basis the latter is preferable.

(c) In this case, the futures contract would lead to a slightly better outcome. This is because it would involve positive cash flows early and negative cash flows later.

(d) In this case, the forward contract would lead to a slightly better outcome. This is because, in the case of the futures contract, the early cash flows would be negative and the later cash flow would be positive.

Tutorial Question

Q7:The following table (also in excel file T2_Q7_data.xls) contains data on monthly changes in the spot price and the futures price for a certain commodity. Use the data to calculate a minimum variance hedge ratio.

Spot Change	Futures Change
0.50	0.56
0.61	0.63
-0.22	-0.12
-0.35	-0.44
0.79	0.60
0.04	-0.06
0.15	0.01
0.70	0.80
-0.51	-0.56
-0.41	-0.46

Figure: Price Change

The standard deviation of the spot (futures) price change: σ_S (σ_F). The correlation coefficient of the price changes: ρ

We can compute these in Excel to find:

$$\sigma_S = 0.4933, \sigma_F = 0.5116, \rho = 0.981$$

Therefore:

$$h^* = \rho \times \frac{\sigma_S}{\sigma_F} = 0.981 \times \frac{0.4933}{0.5116} = 0.946$$

Any other methods?